Cells

Lecture 18

المعهد الهنـدي للتكنولوجيـــا دلهي
أبـــوظبي

INDIAN INSTITUTE OF TECHNOLOGY DELHI
ABU DHABI

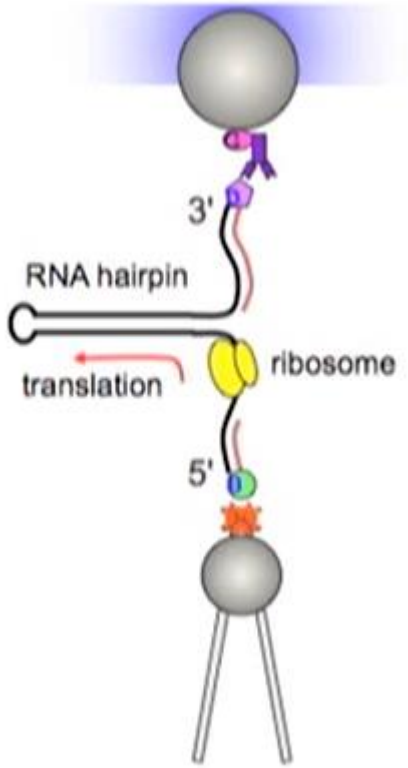
Prof. Saurabh Raj

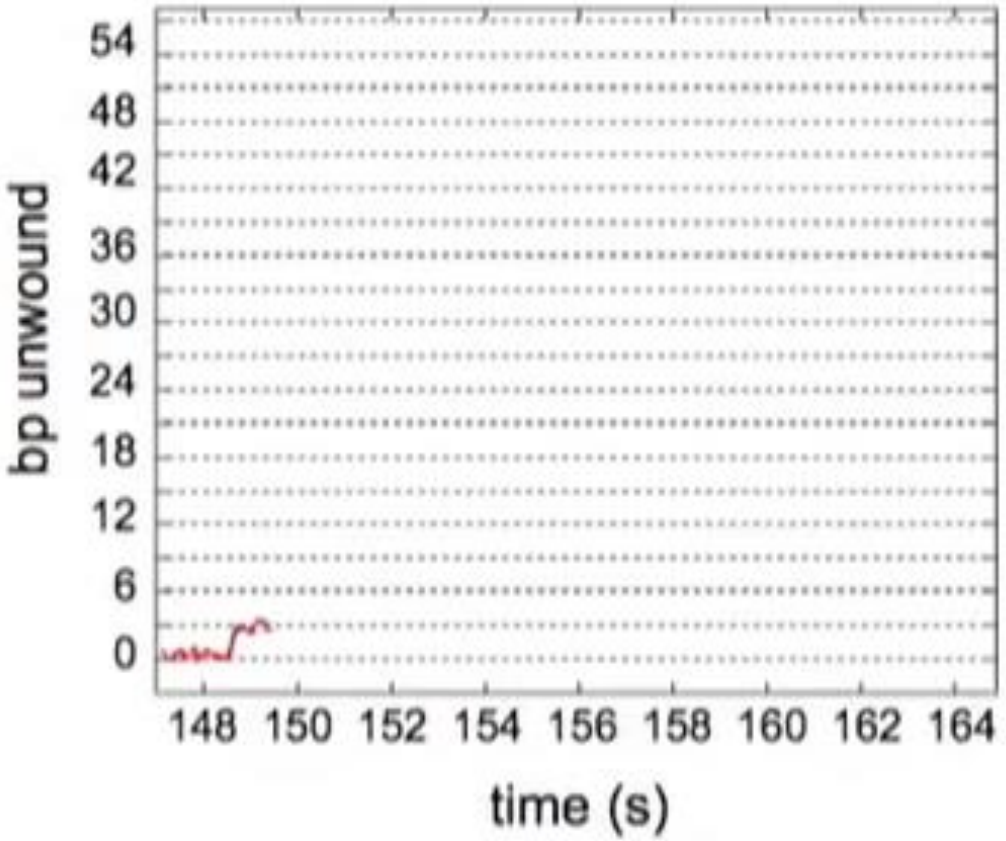
Kusuma School of Biological Sciences
IIT Delhi

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Following real-time translation by ribosome







One can follow translation – one codon at a time!

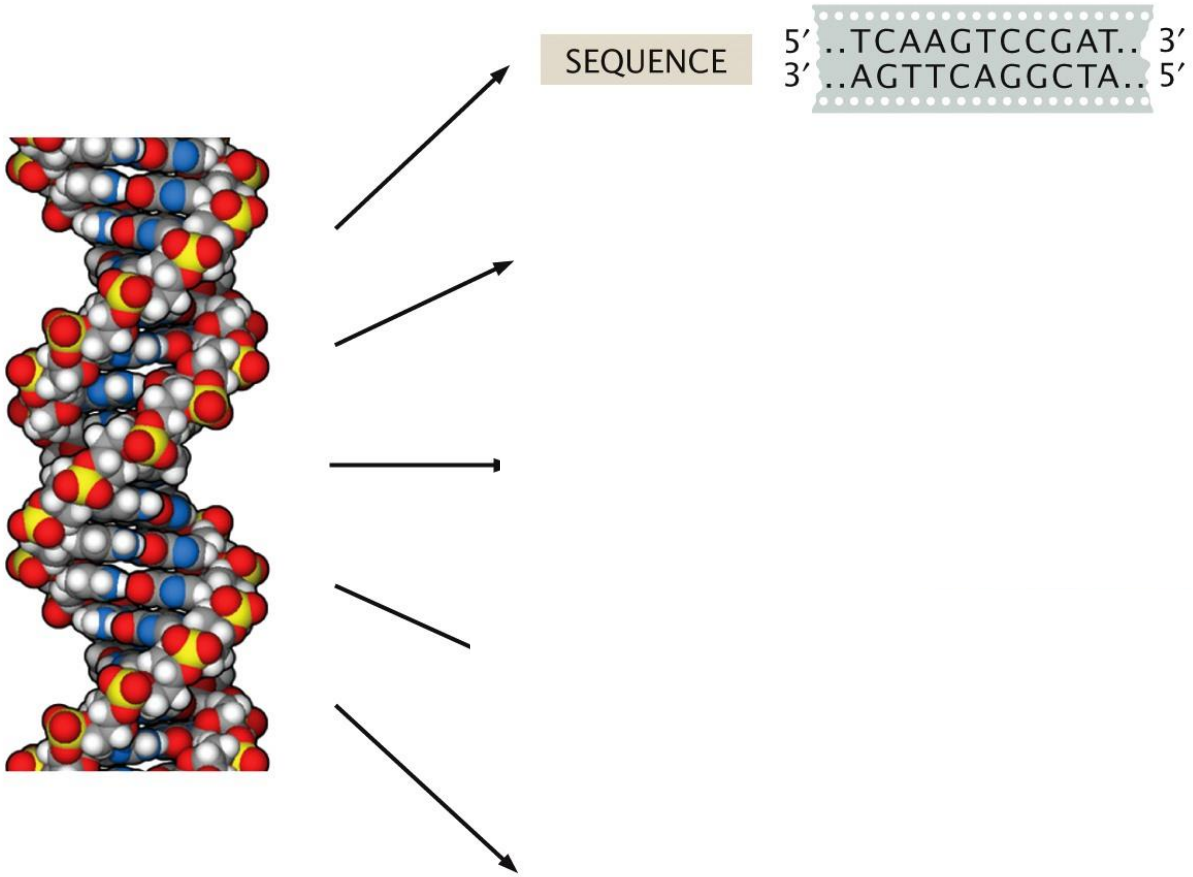
Bustamante's lab



Modelling biomolecules

Idealizations (of DNA)

Biological matter can be idealized using many different physical models



Information content

Figure 1.5 Physical Biology of the Cell, 2ed. (© Garland Science 2013)

Idealizations of a protein

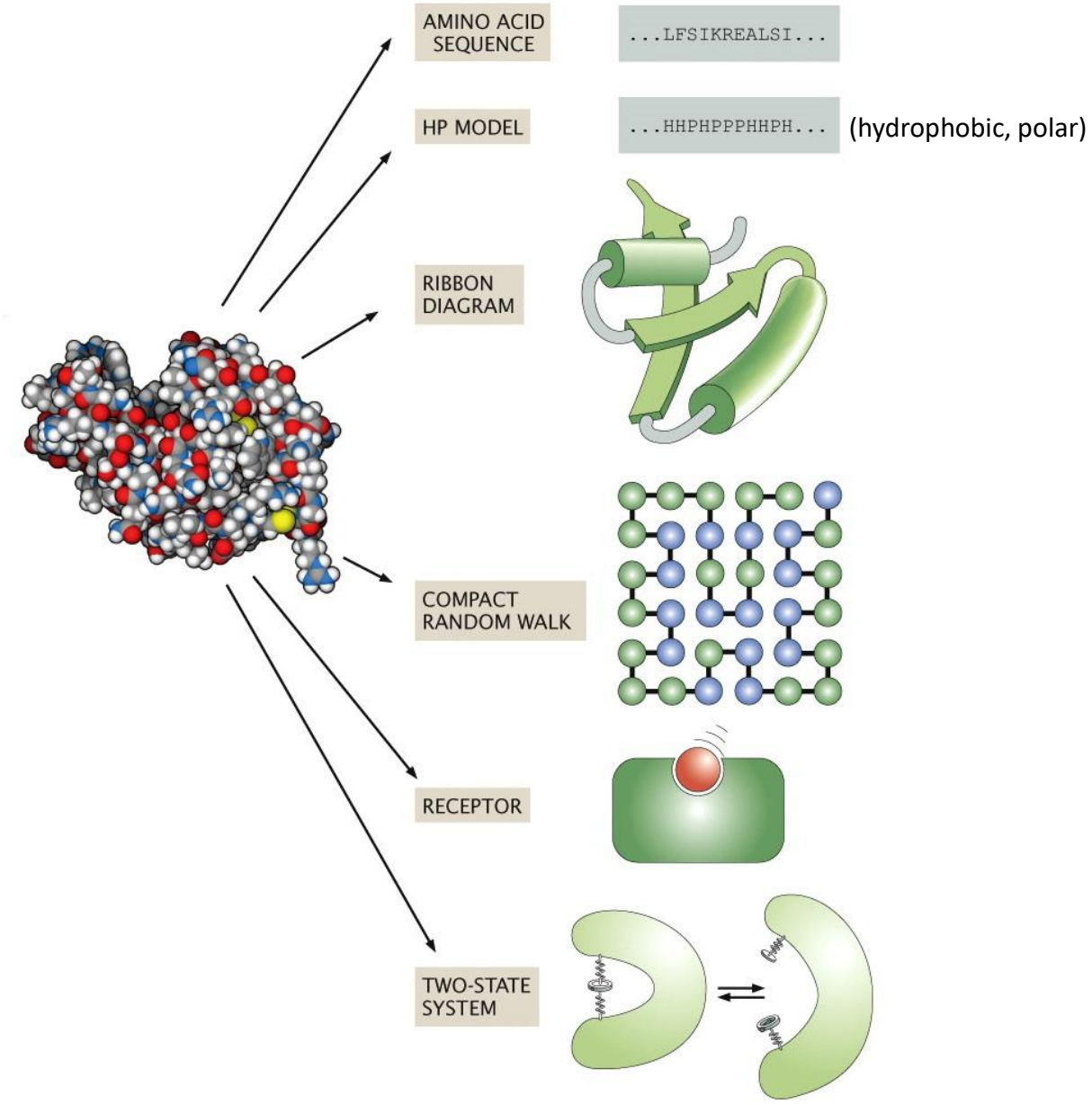
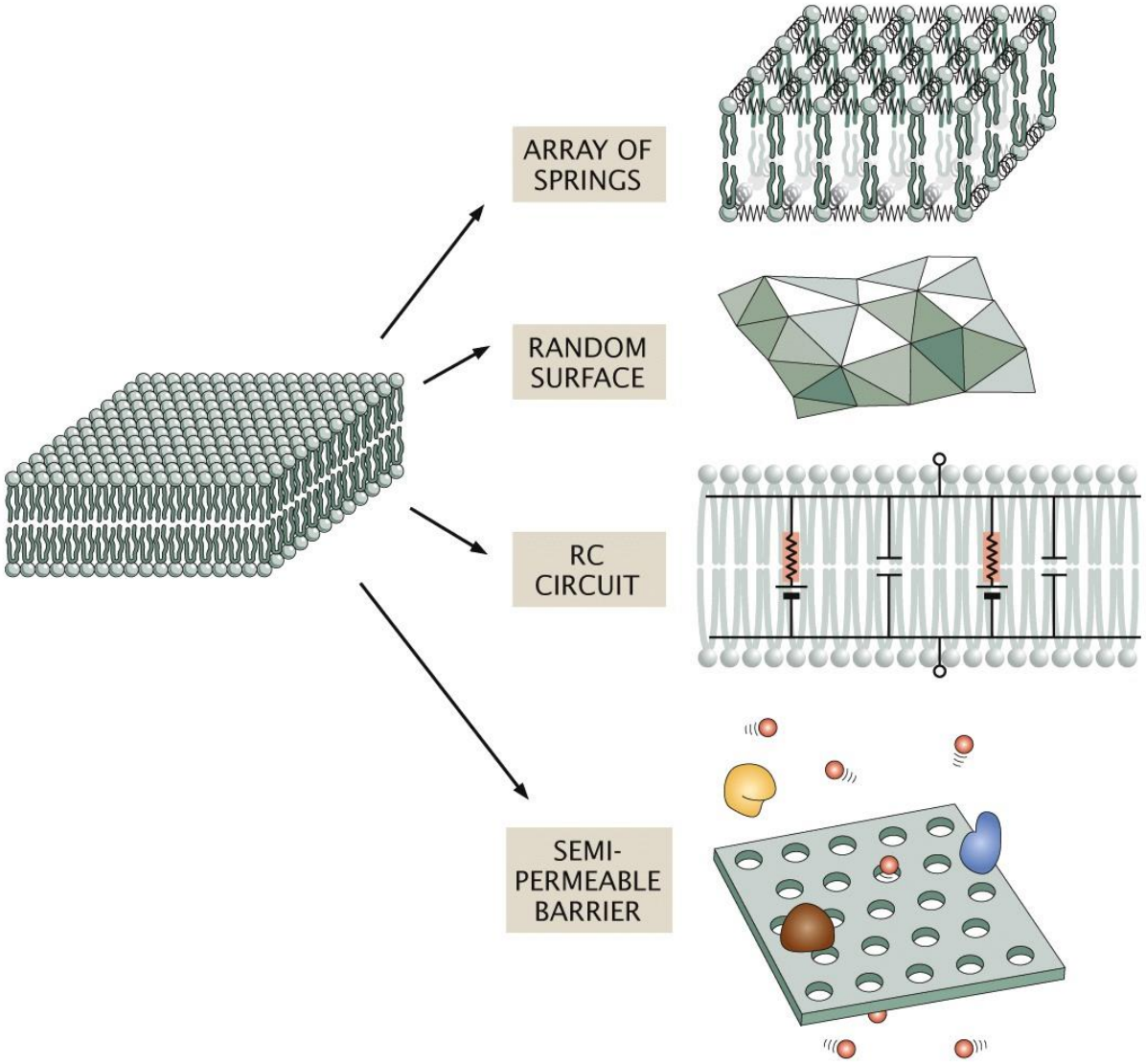


Figure 1.6 Physical Biology of the Cell, 2ed. (© Garland Science 2013)

Idealizations of membranes



membrane potential

Figure 1.7 Physical Biology of the Cell, 2ed. (© Garland Science 2013)

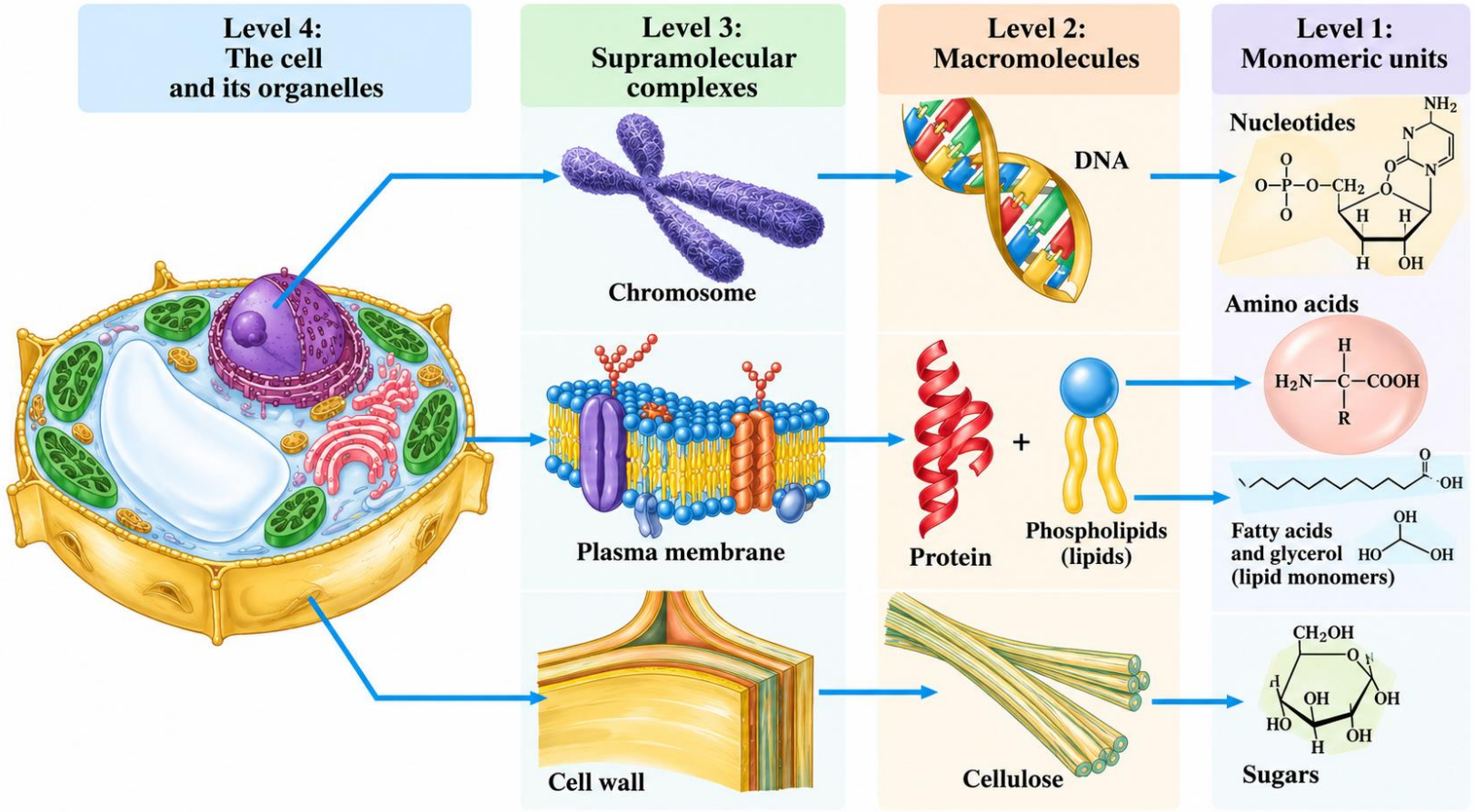


Cells

Fundamental Units of Life

The main biomolecules that are part of the building blocks of cells

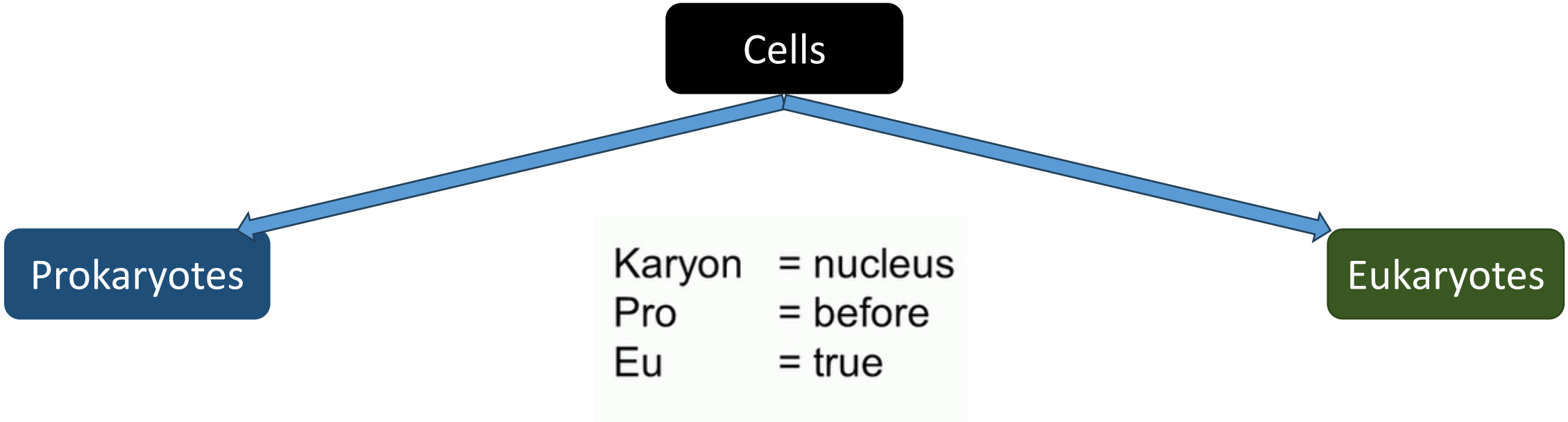
- Carbohydrates and Lipids – cell structure
- Amino acids – proteins
- Nucleotides – DNA and RNA



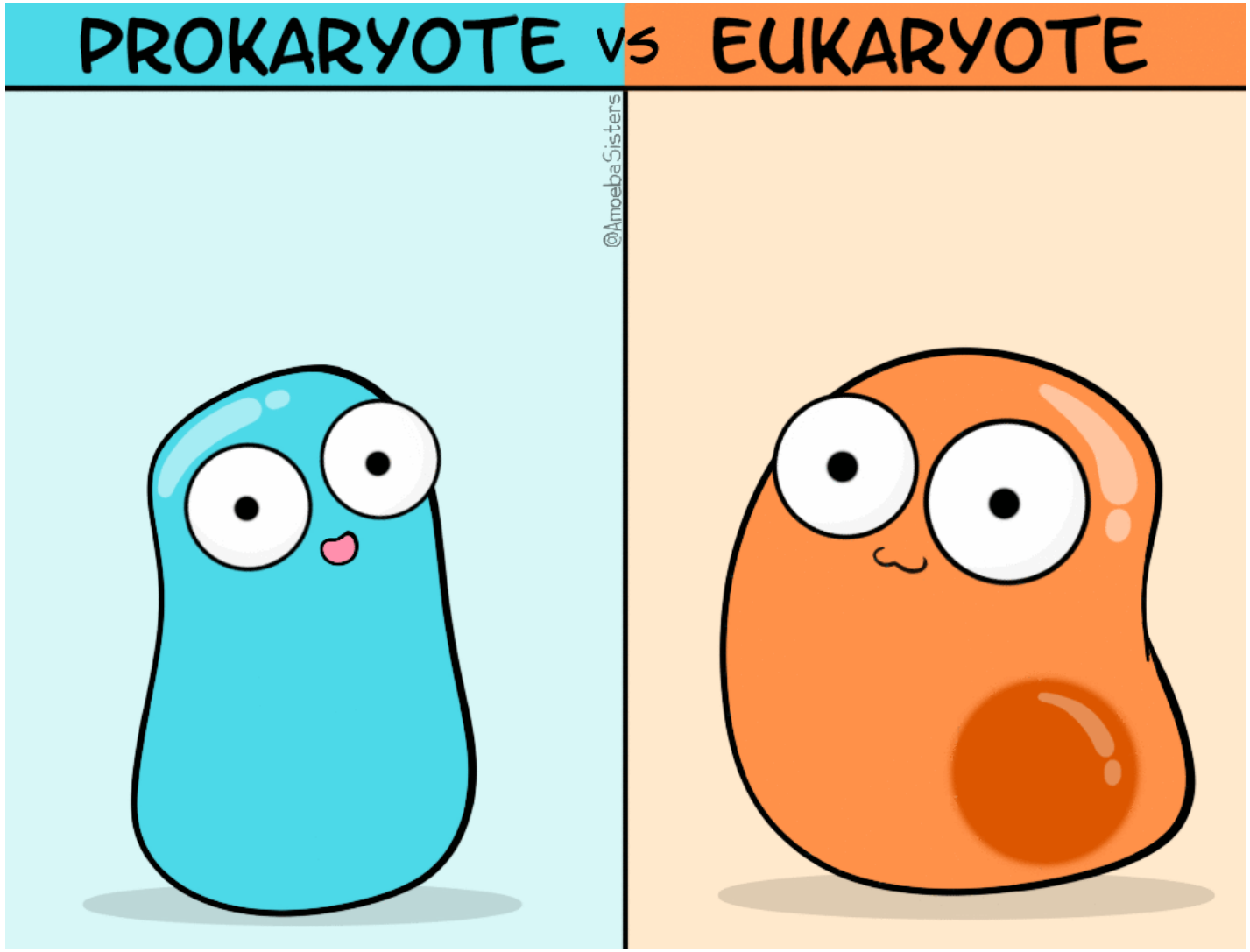


https://www.youtube.com/watch?v=uIGOOcrT86A

Cells – the fundamental unit of life

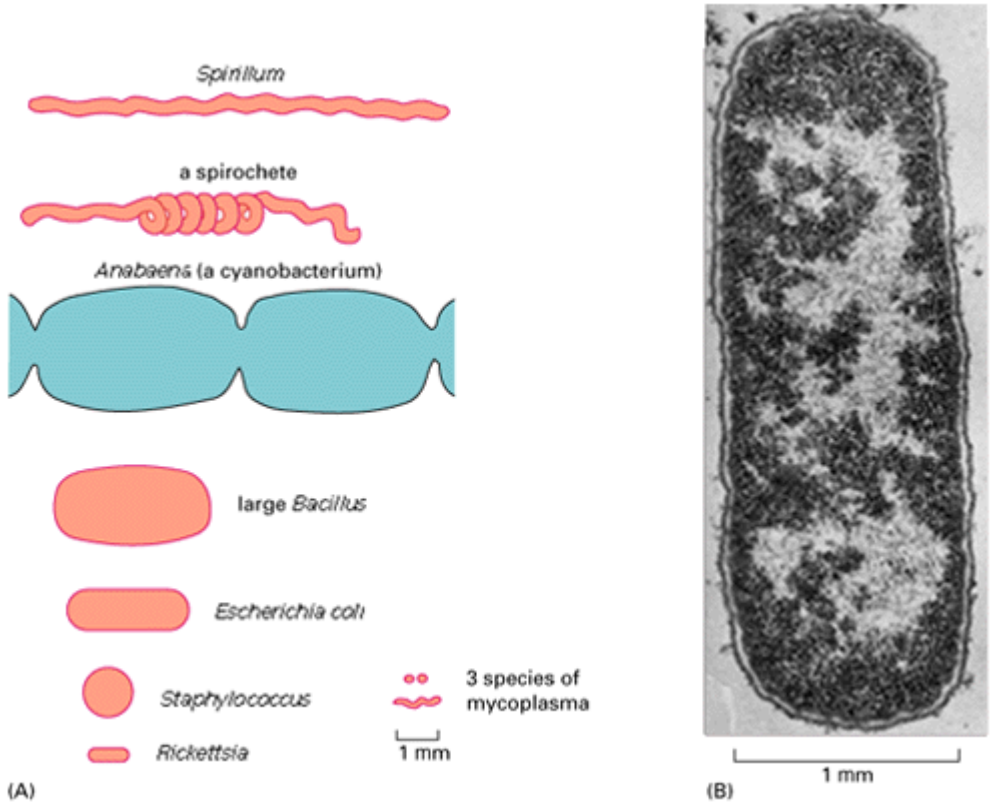


Life is made of cells – either single (archaea, bacteria, protists etc)
or multiple (fungi, animals, plants etc)


PROKARYOTE vs EUKARYOTE
@AmoebaSisters

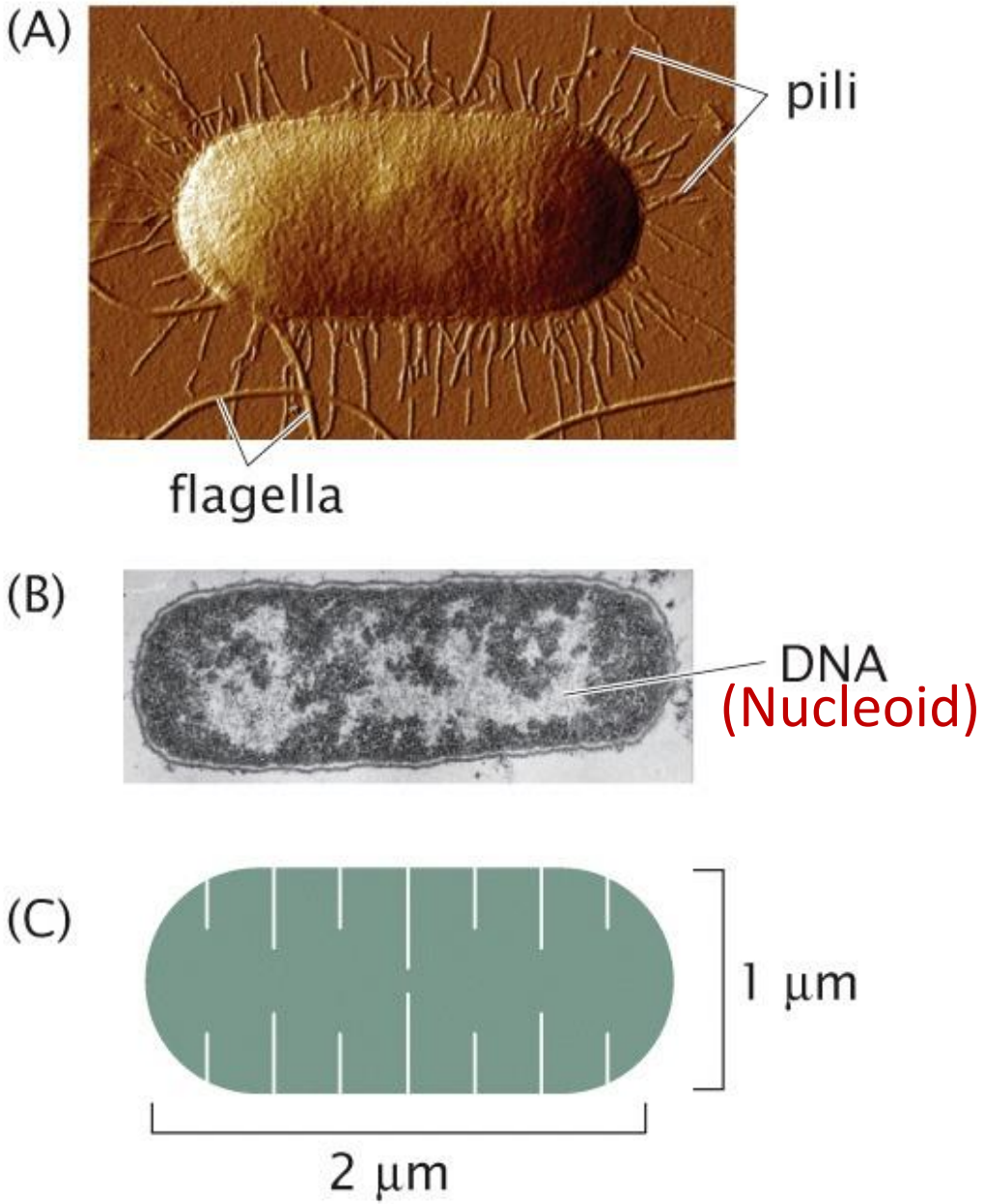
Procaryotic Cells Are Structurally Simple

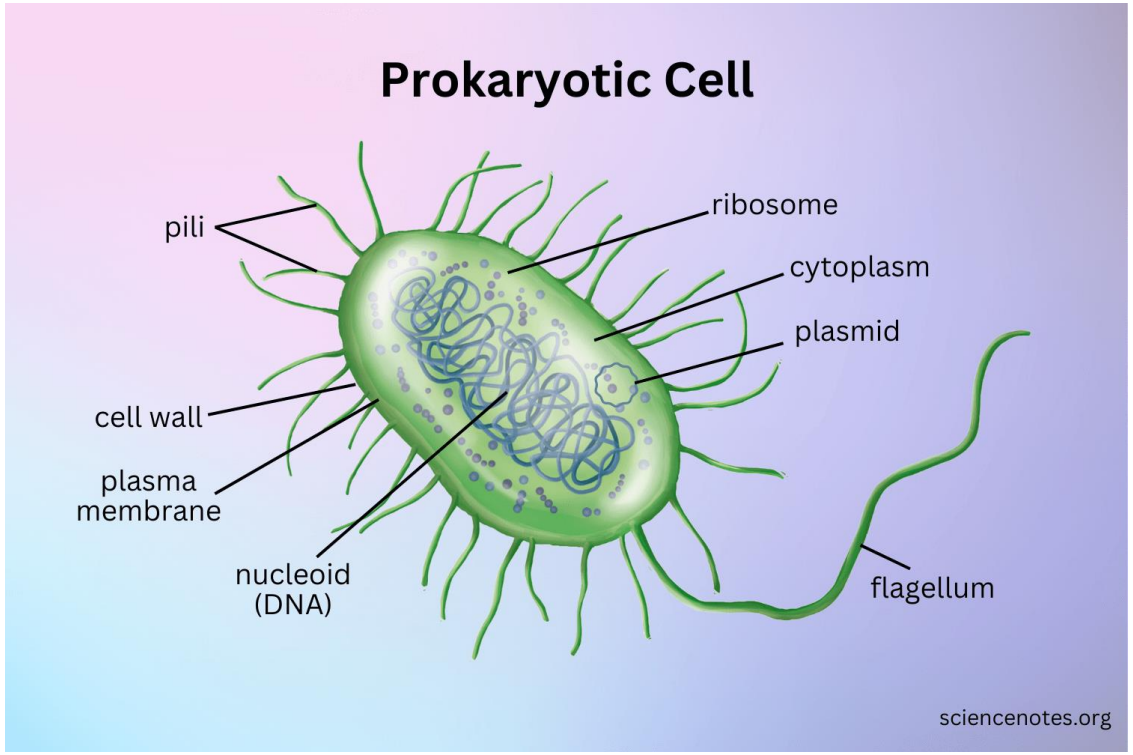
- Bacteria are the simplest organisms found in most natural environments.
- Spherical or rod-shaped cells, commonly several micrometers in linear dimension. Possess a tough protective coat, cell wall, beneath which a plasma membrane encloses a single cytoplasmic compartment containing DNA, RNA, proteins, and small molecules.



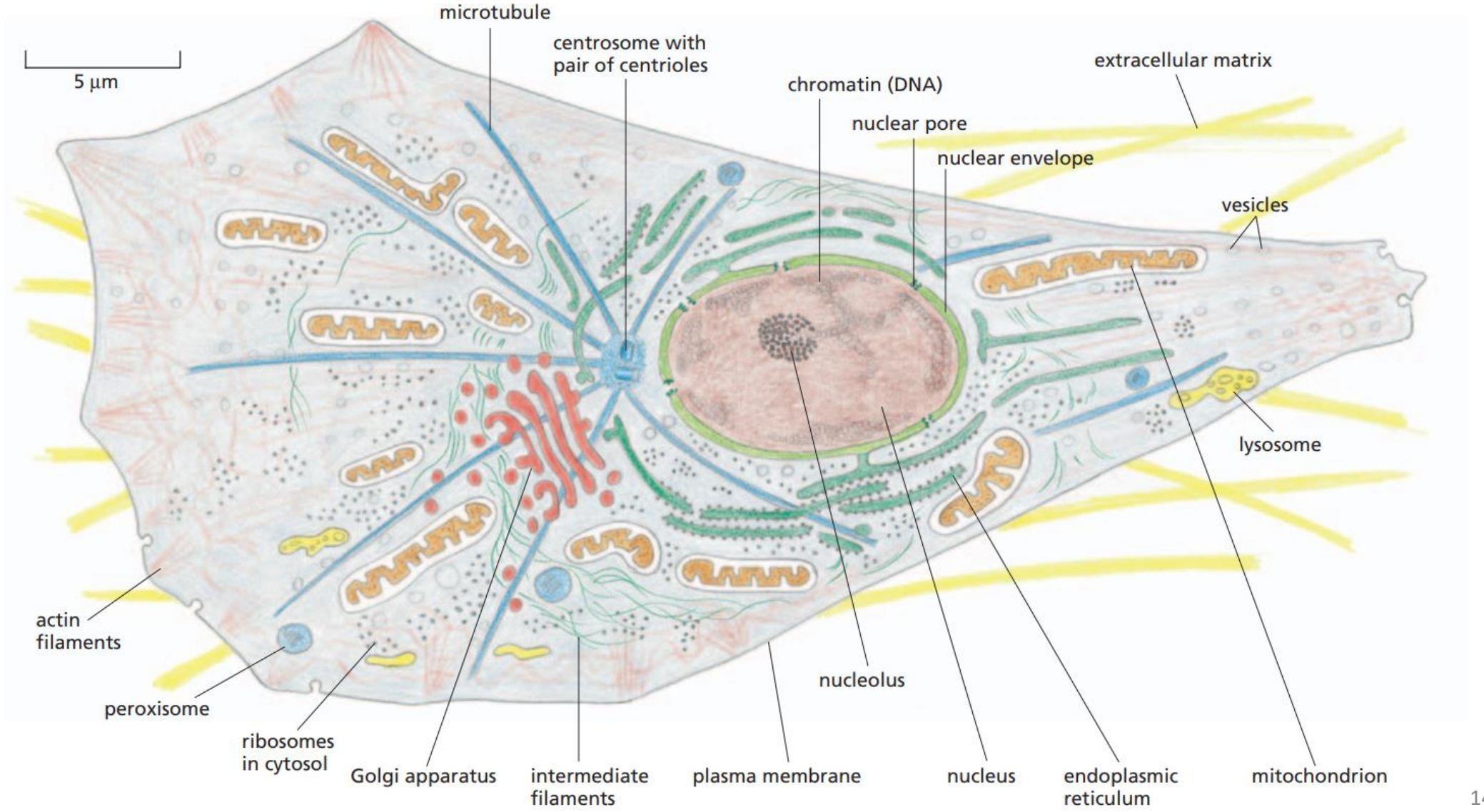
Procaryote sizes and structures. (A) Some procaryotic cells drawn to scale. (B) Electron micrograph of a longitudinal section through a bacterium (Escherichia coli); the cell's DNA is concentrated in the palely stained region. (Courtesy of E. Kellenberger.)

Escherichia coli



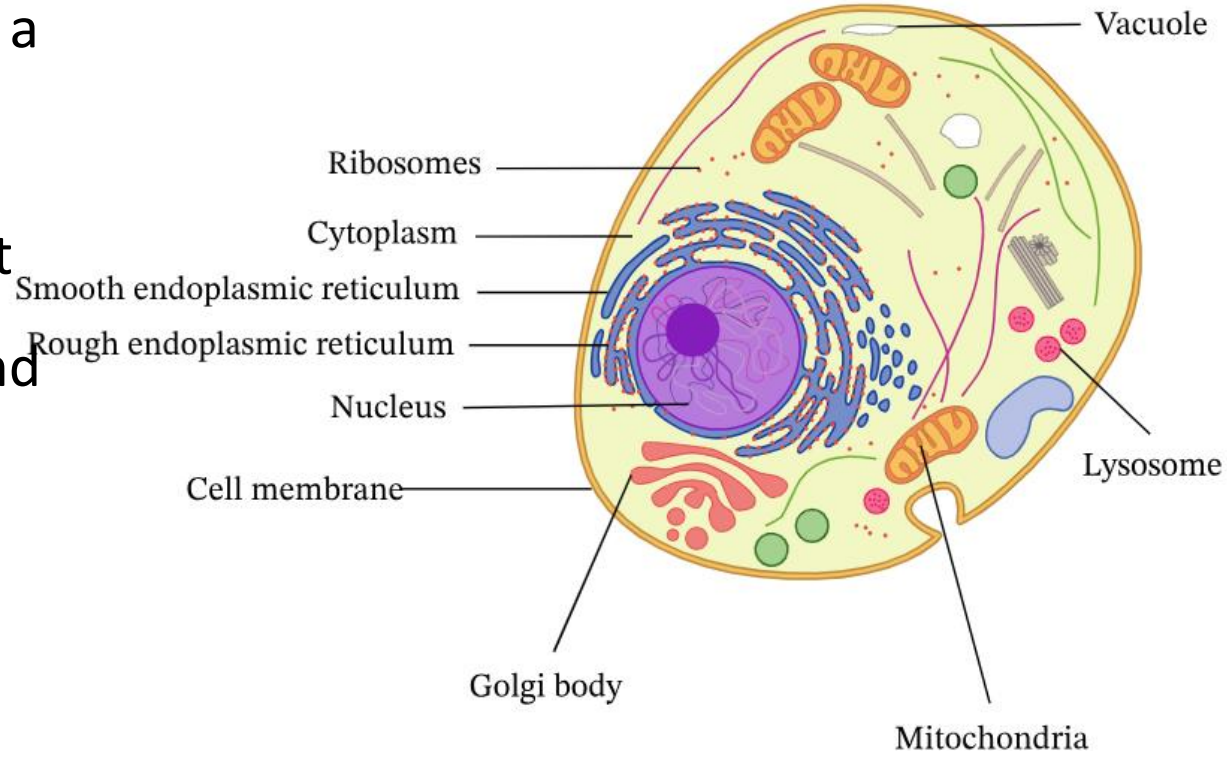


Eukaryotic cells have more complex organization and membrane bound organelle



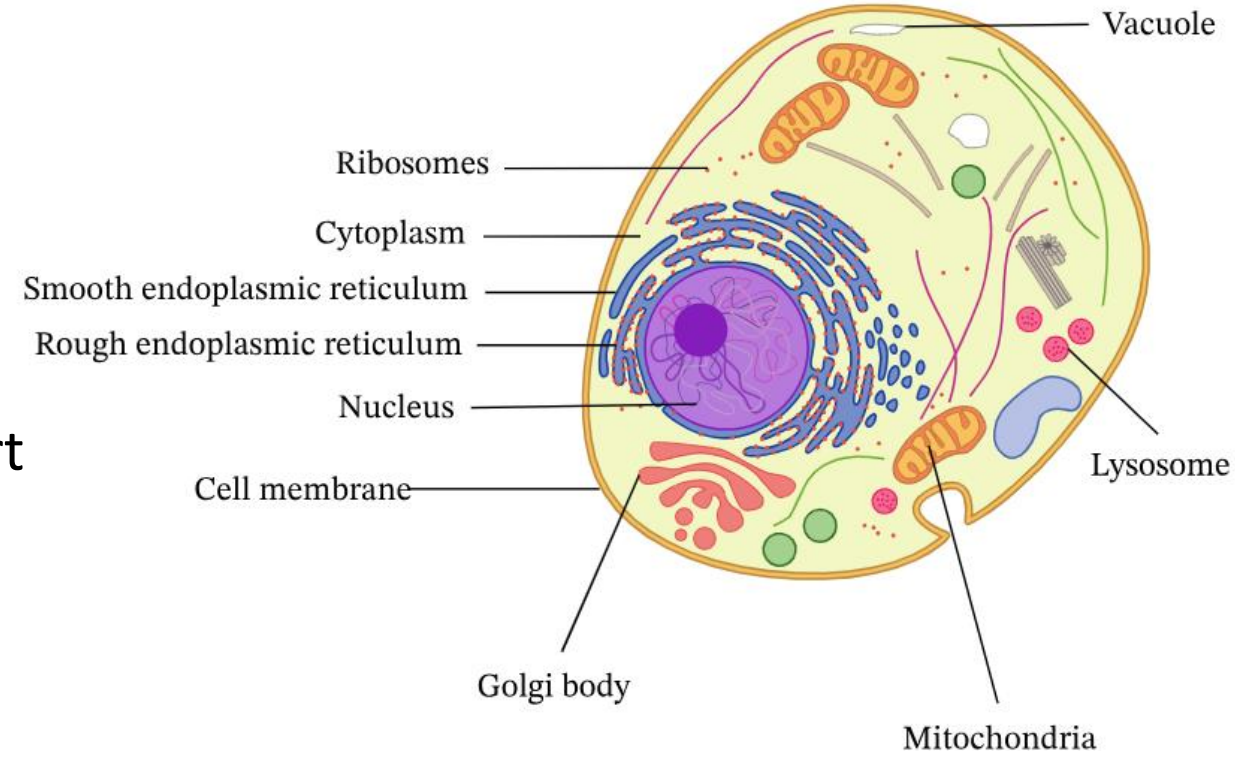
Some common constituent of cells

- **Cell Membrane -** a thin semi-permeable membrane that surrounds the cytoplasm of a cell.
- **Nucleus** - a membrane-bound structure that contains the cell's hereditary information and controls the cell's growth and reproduction.
- **Organelles -** tiny cellular structures, that carry out specific functions necessary for normal cellular operation.



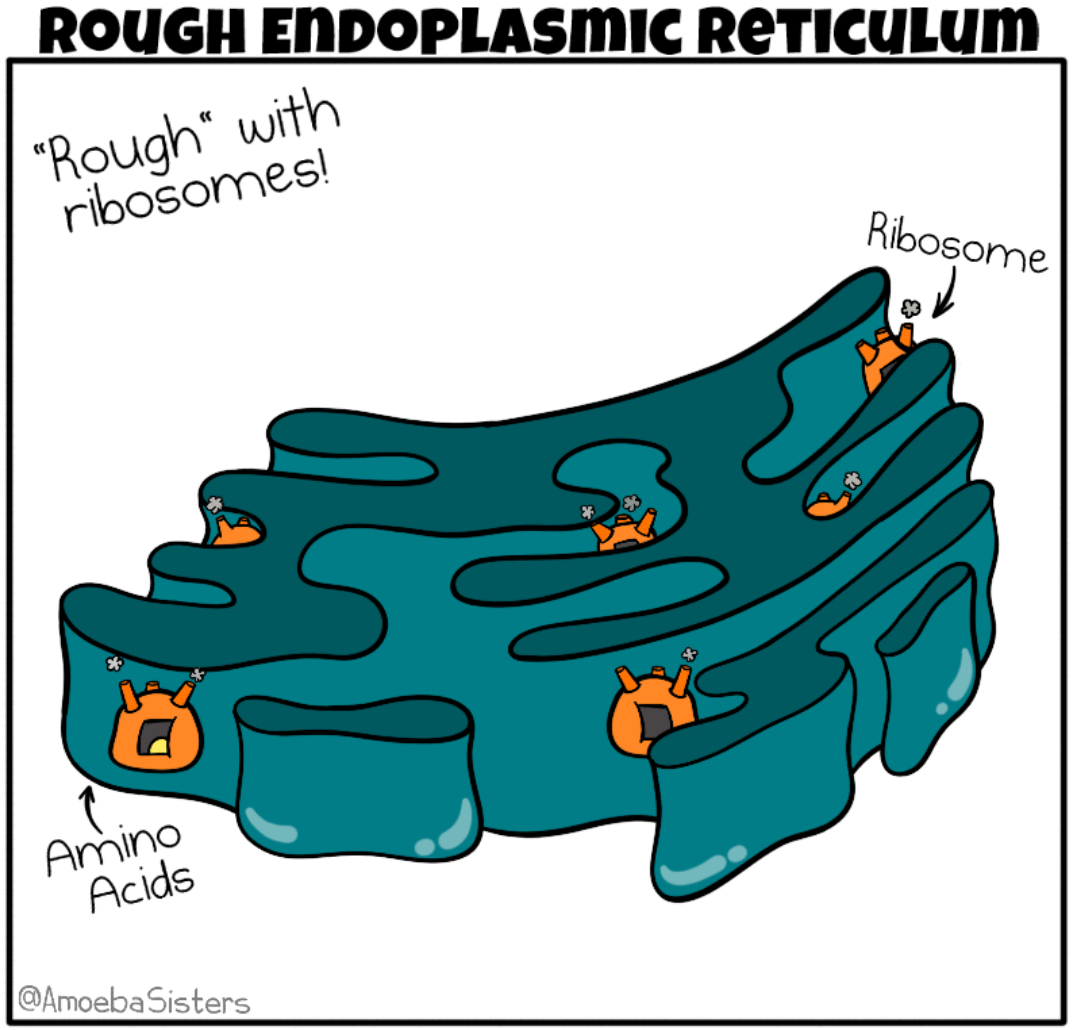
Some common constituent of cells

- **Cytoplasm -** all of the contents outside of the nucleus and enclosed within the cell membrane of a cell.
- **Cytoskeleton -** a network of fibers throughout the cell's cytoplasm that helps the cell maintain its shape and gives support to the cell.
- **Cytosol -** semi-fluid component of a cell's cytoplasm.
- **Mitochondria** - cell organelles that convert energy into forms that are usable by the cell.



Some common constituent of cells

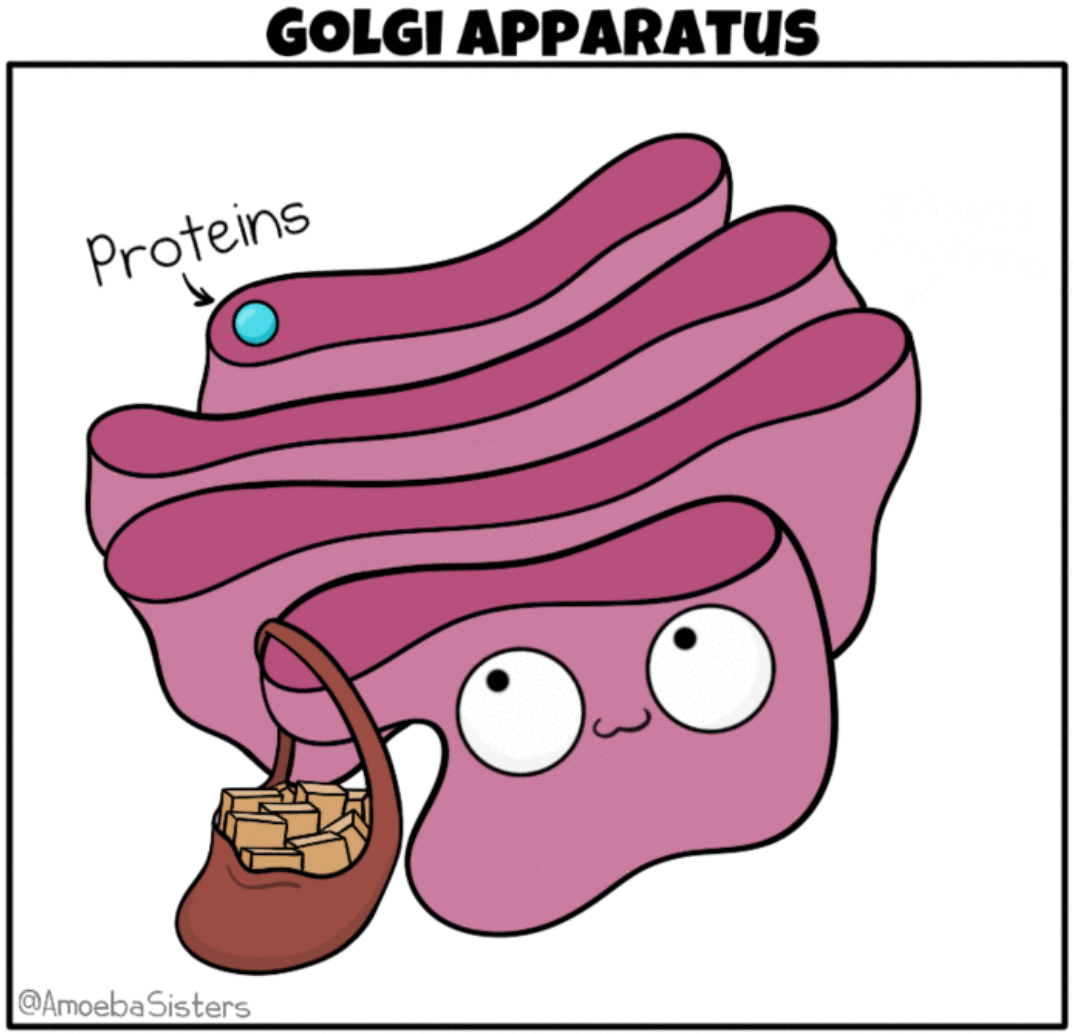
- **Endoplasmic reticulum -** acts as a manufacturing, packaging, and transport system for proteins and lipids.
- **Golgi Body -** acts like a "post office" to further modify, sort, and package proteins and lipids into new vesicles for delivery to their final destination inside or outside the cell.



Protein assembly line of the cell

Some common constituent of cells

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Endosymbiont theory: Mitochondria & Chloroplast derived from Prokaryotic cells

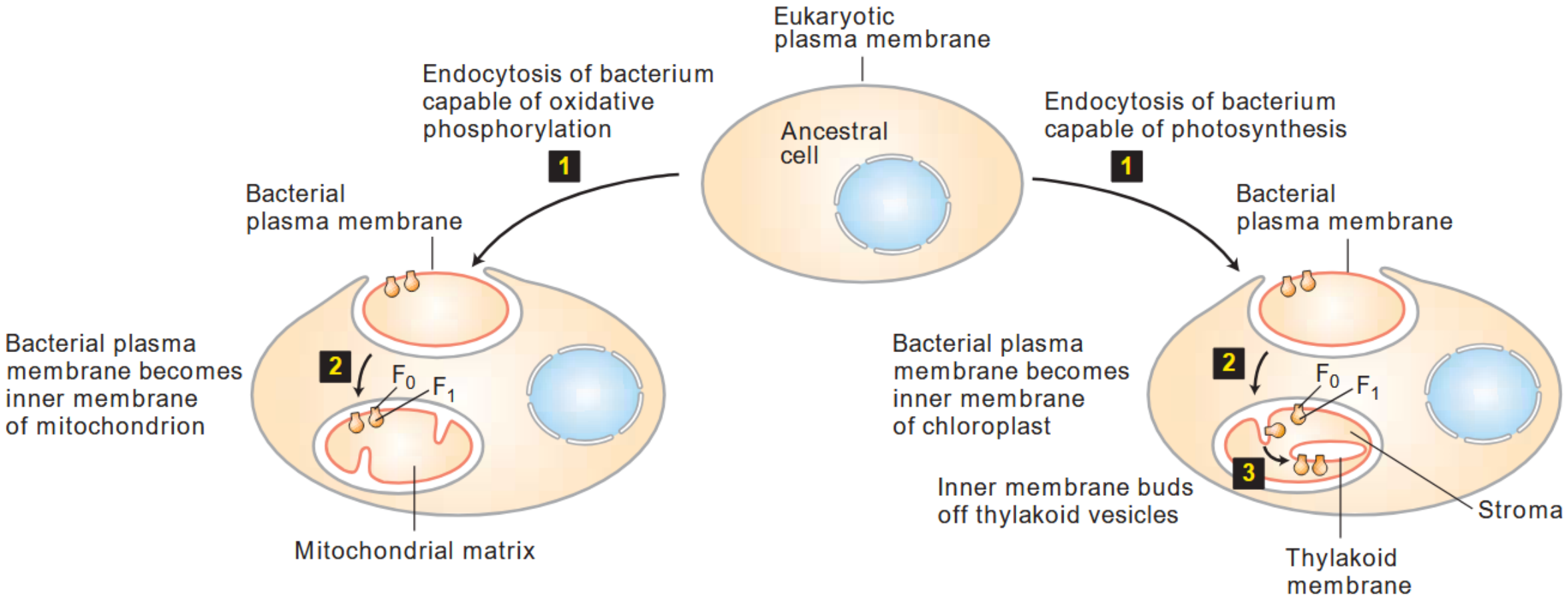
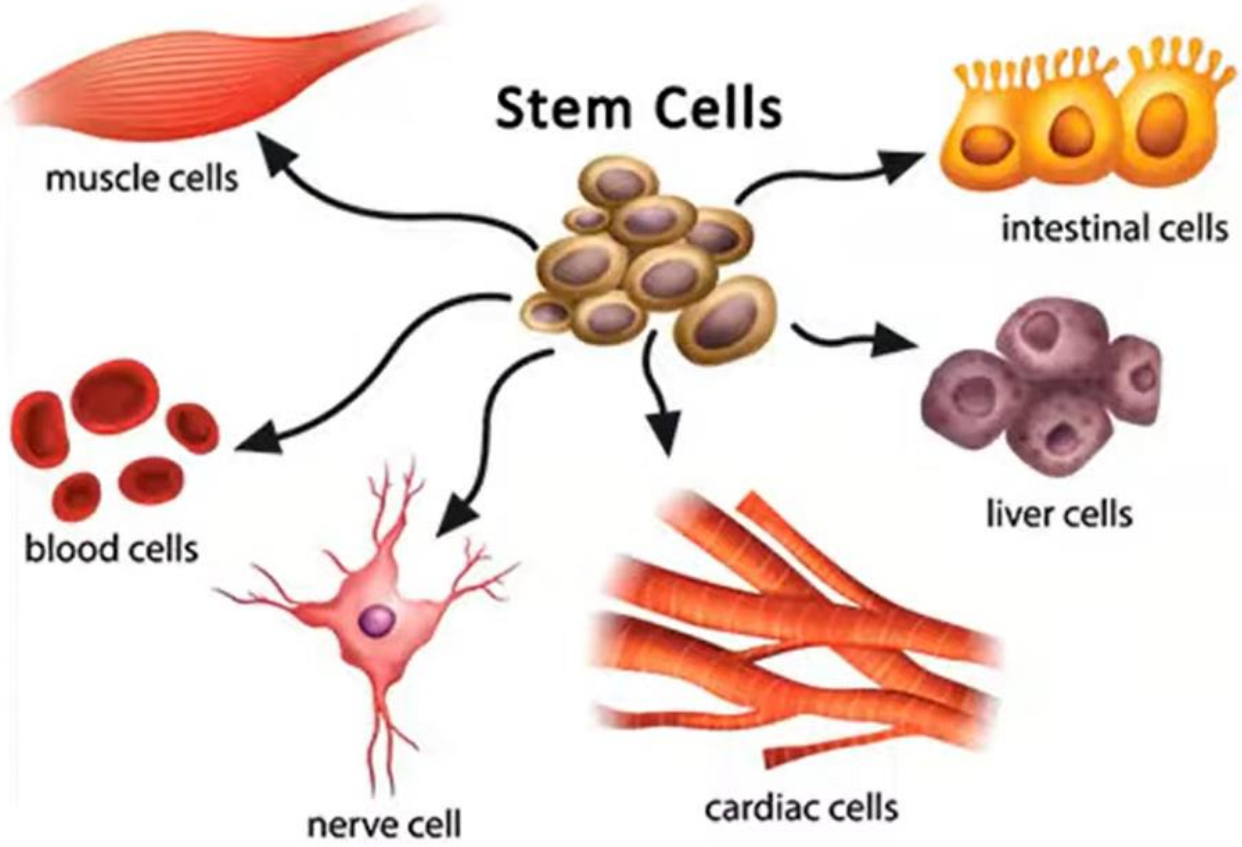


Figure 8-3. Molecular Cell Biology, Freeman Press

Eukaryotes can be highly diverse and specialized

- 50 – 75 Trillion cells in your body
- 220 specialized cells
  - Heart
  - Nerve
  - Blood
  - Sperm
  - Etc.



What renders cellular specialization?

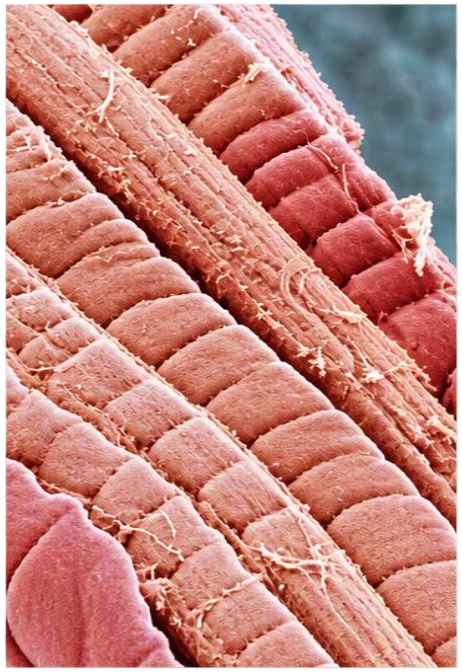
- Most cells have more or less the same genomic DNA.
- What varies is the genes that are being expressed.
- The expressed genes encodes for specific proteins that gives these cells the required specialization.

What are specialized cells?

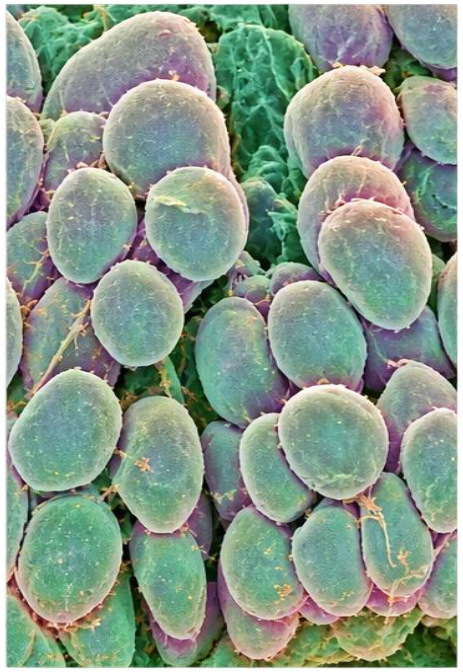
- A cell that has a particular structure and performs a specific function
- Each type has unique shape, size and features allowing it to do its job accurately
- They perform specific functions, such as delivering oxygen, fighting disease, moving the skeleton, transmitting information etc



Red blood cells contain hemoglobin that carries oxygen in blood. The cells are smooth so that they can easily pass through the blood vessels.



Muscle cells are arranged in bundles called muscle fibres. Muscle cells can contract, which makes the fibre shorter and causes bones to move.



Fat cells have a large vacuole in which to store fat molecules. This is how the cell stores chemical energy.



Nerve cells are long, thin, and have many branches. They conduct electrical impulses to coordinate body activity.

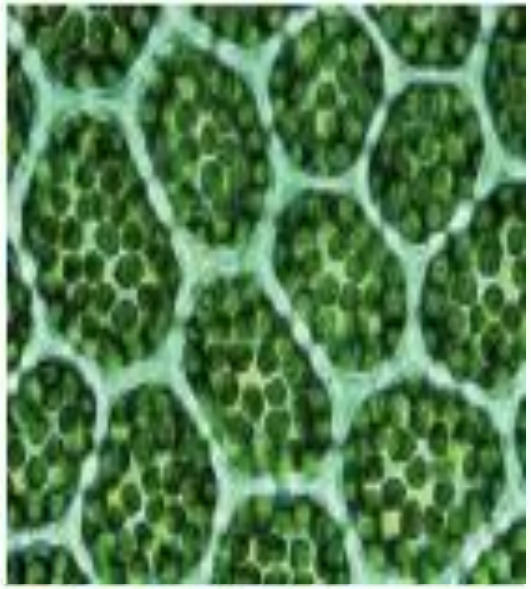
Plant cells also have a variety of specialized cells. Cells in the leaf of a tree have a different structure and function from the cells in the trunk.



Epidermal cells on young roots have hairs that absorb water from soil



Some plant cells transport water and minerals throughout the plant



Photosynthetic cells contain many chloroplasts to collect energy from sun to make sugar for the plant



Some cells transport dissolved sugars around the plant